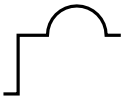
Math Kangaroo Practice

Grade 3-4 · 20 questions · standard · seed 2026

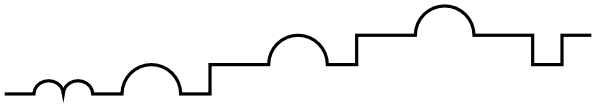
Part A: Each question is worth 3 points

1. Lena cut a strip of paper along this line.

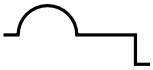
Which piece **cannot** come from the cut line?

A. ***B.*** ***C.*** ***D.***  ***E.***

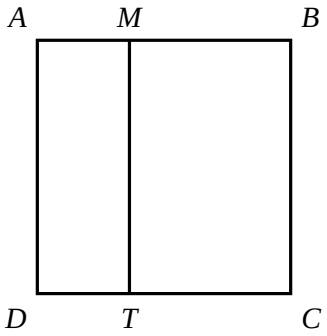
2. Far away we see the skyline of a castle.



Which of the pieces **cannot** belong to the skyline?

A. ***B.*** ***C.*** ***D.***  ***E.***

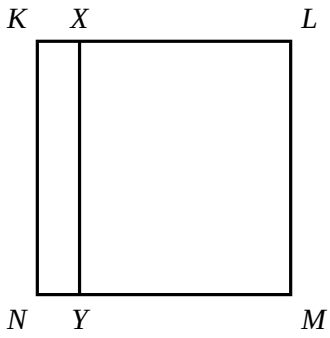
3. *ABCD* is a square. Its side is 11 cm. The line *MT* cuts it into two rectangles, and *AM* = 4 cm.



How many centimetres is the perimeter of the square *ABCD* larger than that of the rectangle *MBCT*?

A. 36 cm ***B.*** 22 cm ***C.*** 16 cm ***D.*** 14 cm ***E.*** 8 cm

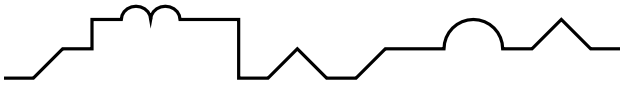
4. *KLMN* is a square. Its side is 12 cm. *KXYN* is a rectangle. Its shorter side is 2 cm.



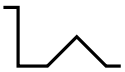
How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *KXYN*?

A. 20 cm ***B.*** 19 cm ***C.*** 15 cm ***D.*** 10 cm ***E.*** 4 cm

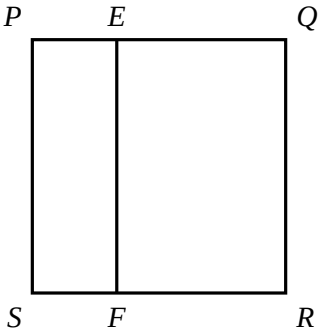
5. This is the outline of a town seen from the river at sunset.



Which piece **cannot** be part of the outline?

A.  ***B.*** ***C.*** ***D.*** ***E.***

6. *PQRS* is a square. Its side is 6 cm. *PEFS* is a rectangle. Its shorter side is 2 cm.

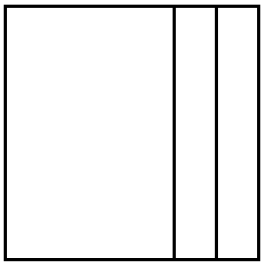


How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *PEFS*?

A. 4 cm ***B.*** 5 cm ***C.*** 6 cm ***D.*** 7 cm ***E.*** 8 cm

Part B: Each question is worth 4 points

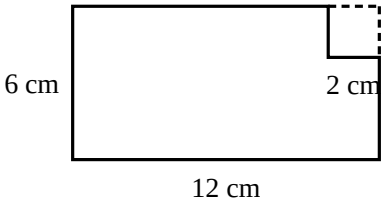
7. A square has perimeter 20 cm. It is cut along straight lines into 3 rectangles as shown.



What is the sum of the perimeters of all 3 rectangles?

A. 20 cm ***B.*** 40 cm ***C.*** 41 cm ***D.*** 50 cm ***E.*** 60 cm

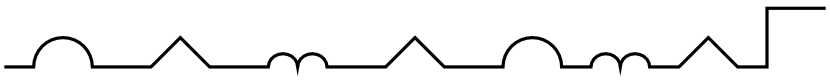
8. A rectangle is 12 cm long and 6 cm wide. A square with side 2 cm is cut off one corner.



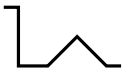
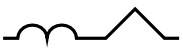
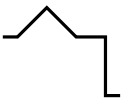
What is the perimeter of the remaining shape?

A. 28 cm ***B.*** 32 cm ***C.*** 34 cm ***D.*** 36 cm ***E.*** 40 cm

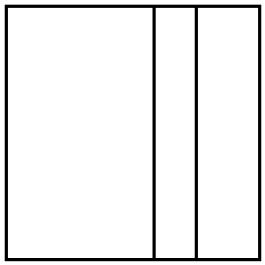
9. This is the outline of a town seen from the river at sunset.



Which piece **can** be part of the outline?

A. ***B.***  ***C.***  ***D.*** ***E.*** 

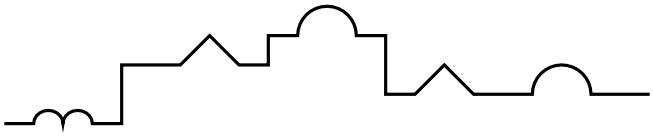
10. Grandma bakes a square cake. A ribbon 48 cm long fits exactly around it. She cuts the cake into 3 rectangular pieces as shown and puts a ribbon around every piece.



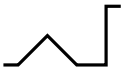
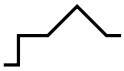
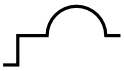
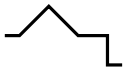
How many centimetres of ribbon does she need for all the pieces together?

A. 144 cm ***B.*** 111 cm ***C.*** 104 cm ***D.*** 96 cm ***E.*** 72 cm

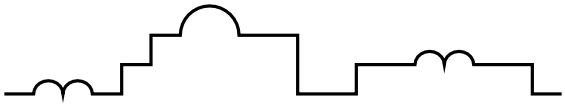
11. Lena cut a strip of paper along this line.



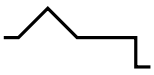
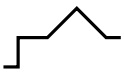
Which piece **can** come from the cut line?

A.  ***B.***  ***C.***  ***D.*** ***E.*** 

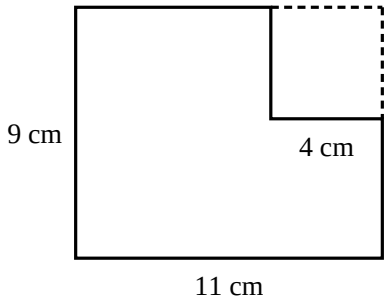
12. Far away we see the skyline of a castle.



Which of the pieces **can** belong to the skyline?

A. ***B.***  ***C.*** ***D.***  ***E.***

13. A rectangle is 11 cm long and 9 cm wide. A square with side 4 cm is cut off one corner.

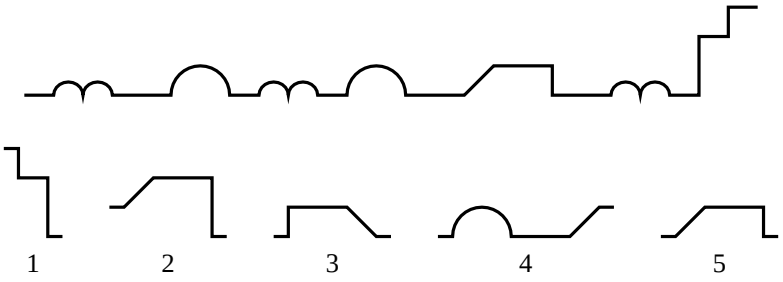


What is the perimeter of the remaining shape?

A. 36 cm ***B.*** 40 cm ***C.*** 47 cm ***D.*** 48 cm ***E.*** 56 cm

Part C: Each question is worth 5 points

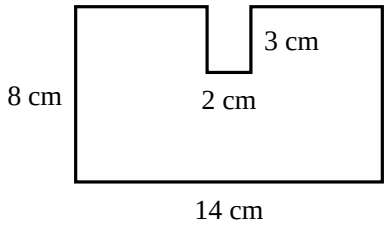
14. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

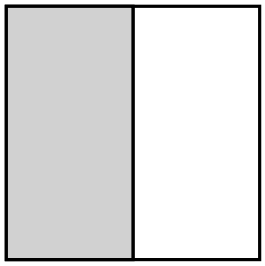
15. A 14 cm by 8 cm rectangle of cardboard has a rectangular notch 2 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 50 cm ***B.*** 54 cm ***C.*** 55 cm ***D.*** 60 cm ***E.*** 62 cm

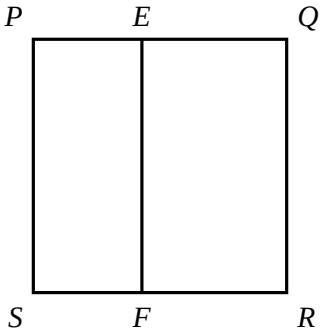
16. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 18 cm.



How long is the side of the square?

A. 2 cm ***B.*** 3 cm ***C.*** 4 cm ***D.*** 5 cm ***E.*** 6 cm

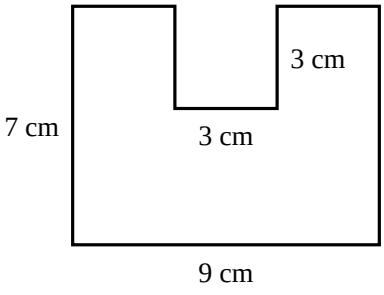
17. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 20 cm and the perimeter of *EQRF* is 22 cm.



How long is the side of the square *PQRS*?

A. 7 cm ***B.*** 10 cm ***C.*** 14 cm ***D.*** 28 cm ***E.*** 35 cm

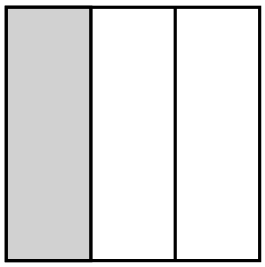
18. A 9 cm by 7 cm rectangle of cardboard has a rectangular notch 3 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 44 cm ***B.*** 43 cm ***C.*** 42 cm ***D.*** 40 cm ***E.*** 38 cm

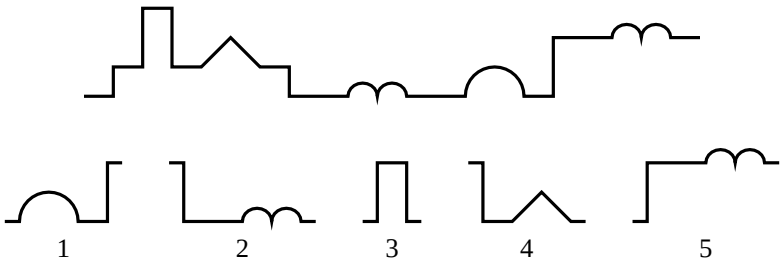
19. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 48 cm.



What is the perimeter of the square?

A. 18 cm ***B.*** 36 cm ***C.*** 60 cm ***D.*** 71 cm ***E.*** 72 cm

20. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

Answer Key

1.	**E**	**2.**	**C**	**3.**	**E**	**4.**	**A**	**5.**	**D**	**6.**	**E**
7.	**B**	**8.**	**D**	**9.**	**C**	**10.**	**D**	**11.**	**C**	**12.**	**C**
13.	**B**	**14.**	**C**	**15.**	**A**	**16.**	**E**	**17.**	**A**	**18.**	**E**
19.	**E**	**20.**	**D**								

Solutions

1. E — All the other pieces can be found in the line, but this one cannot: it is a mirror image of a part of the line. [2002 Q7 · cannot_belong]

2. C — All the other pieces can be found in the line, but this one cannot: its flat part between the two features has the wrong length. [2002 Q7 · cannot_belong]

3. E (8 cm) — *MBCT* has sides 7 and 11 cm, perimeter 36 cm. Square: 44 cm. Difference 8 cm (= 2 × 4). [2002 Q6 · square_minus_strip]

4. A (20 cm) — Square: 4 × 12 = 48 cm. Rectangle: 2 + 12 + 2 + 12 = 28 cm. Difference 48 – 28 = 20 cm. [2002 Q6 · square_minus_strip]

5. D — All the other pieces can be found in the line, but this one cannot: its step is taller than in the line. [2002 Q7 · cannot_belong]

6. E (8 cm) — Square: 4 × 6 = 24 cm. Rectangle: 2 + 6 + 2 + 6 = 16 cm. Difference 24 – 16 = 8 cm. [2002 Q6 · square_minus_strip]

7. B (40 cm) — Side = 20 ÷ 4 = 5 cm. There are 2 cuts; each cut is 5 cm long and is counted twice (once for each piece next to it): 20 + 2 × 2 × 5 = 40 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

8. D (36 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (12 + 6) = 36 cm. [2002 Q6 · corner_cut]

9. C — Only this piece is found in the line. The others don't fit — A: it has two small bumps where the line has a pointed roof; B: it is a mirror image of a part of the line; D: it has a round dome where the line has a pointed roof; E: it steps down where the line steps up. [2002 Q7 · can_belong]

10. D (96 cm) — Side = 48 ÷ 4 = 12 cm. There are 2 cuts; each cut is 12 cm long and is counted twice (once for each piece next to it): 48 + 2 × 2 × 12 = 96 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

11. C — Only this piece is found in the line. The others don't fit — A: it is a mirror image of a part of the line; B: it has a pointed roof where the line has a round dome; D: it steps down where the line steps up; E: it steps down where the line steps up. [2002 Q7 · can_belong]

12. C — Only this piece is found in the line. The others don't fit — A: it steps down where the line steps up; B: it has a pointed roof where the line has two small bumps; D: it has a pointed roof where the line has a round dome; E: it is a mirror image of a part of the line. [2002 Q7 · can_belong]

13. B (40 cm) — The two new edges (4 cm and 4 cm) replace the two removed pieces of the outer edge, which were also 4 cm and 4 cm. So the perimeter is the same as the rectangle's: 2 × (11 + 9) = 40 cm. [2002 Q6 · corner_cut]

14. C (2) — Pieces 4, 5 fit. Not: 1 (it is a mirror image of a part of the line); 2 (its step is taller than in the line); 3 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

15. A (50 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (14 + 8) + 2 × 3 = 50 cm. [2002 Q6 · notch]

16. E (6 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 18 ÷ 6 = 3, so the side is 2 × 3 = 6 cm. [2002 Q6 · strips_to_square]

17. A (7 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 20 + 22 = 42 cm = 6 sides. Side = 42 ÷ 6 = 7 cm. [2002 Q6 · two_perimeters]

18. E (38 cm) — The bottom of the notch (3 cm) replaces the 3 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (9 + 7) + 2 × 3 = 38 cm. [2002 Q6 · notch]

19. E (72 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 48 ÷ 8 = 6, so the side is 3 × 6 = 18 cm and the perimeter is 72 cm. [2002 Q6 · strips_to_square]

20. D (4) — Pieces 1, 3, 4, 5 fit. Not: 2 (it steps down where the line steps up). [2002 Q7 · how_many_fit]